CCNA NETWORKING INTERNSHIP

WEEK 1 – TASK 1

Build Your CCNA Lab + How Networks Actually Work

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1. Introduction

The purpose of this task is to understand the basic working of computer networks and gain practical experience using Cisco Packet Tracer.

In this lab, I designed a small network consisting of two routers, two switches, and four PCs. I configured IP addresses, default gateways, and static routing to establish communication between all four computers.

This practical helped me understand how network devices communicate and how data travels from one computer to another.

2. Objectives

The main objectives of this task are:

1. Create a Cisco Networking Academy account.
2. Install and explore Cisco Packet Tracer.
3. Build a network topology using two routers, two switches, and four PCs.
4. Configure IP addresses on all network devices.
5. Establish communication between all four PCs.
6. Understand the functions of routers, switches, and hubs.
7. Trace the path of a ping packet from PC1 to PC4.

3. Tools and Software Used

Tool	Purpose
Cisco Packet Tracer	Network simulation
Cisco Networking Academy	Learning resources
2 Routers	Communication between networks
2 Switches	Connecting computers
4 PCs	End devices for testing
Copper Straight-Through	Connecting PCs to switches and switches to routers
Copper Cross-Over	Connecting routers directly

4. Network Topology

The network consists of two separate LANs connected through two routers.

- LAN 1 contains PC1, PC2, and Switch 1.
- LAN 2 contains PC3, PC4, and Switch 2.
- Router 1 and Router 2 connect both networks.
- Static routing is configured to allow communication between the two LANs.

Figure 1: Complete Network Topology

[INSERT YOUR PACKET TRACER TOPOLOGY SCREENSHOT HERE]

5. Network Devices and Their Functions

5.1 Router

A router connects different networks and forwards packets between them using IP addresses.

In my topology, two routers are used to connect LAN 1 and LAN 2. They allow computers on different networks to communicate with each other.

5.2 Switch

A switch connects multiple devices within the same local area network (LAN).

It uses MAC addresses to forward Ethernet frames to the correct destination device.

In this lab, each switch connects two PCs and one router.

5.3 Hub

A hub is a basic networking device that connects multiple computers. Unlike a switch, it sends incoming data to all connected ports instead of forwarding it only to the destination port.

A hub does not intelligently filter traffic using MAC addresses.

5.4 Personal Computers (PCs)

PCs are end devices used to send and receive network data.

In this lab, four PCs are configured with IP addresses to test communication between the two LANs.

6. IP Addressing Table

The following IP addressing scheme is used in the lab.

Device	Interface	IP Address	Subnet Mask
PC1	FastEthernet0	192.168.10.2	255.255.255.0
PC2	FastEthernet0	192.168.10.3	255.255.255.0
Router 1	FastEthernet0/0	192.168.10.1	255.255.255.0
Router 1	Serial0/3/0	10.0.0.1	255.255.255.252
Router 2	Serial0/3/0	10.0.0.2	255.255.255.252
Router 2	FastEthernet0/0	192.168.20.1	255.255.255.0

PC3	FastEthernet0	192.168.20.2	255.255.255.0
PC4	FastEthernet0	192.168.20.3	255.255.255.0

Default Gateways

- PC1 and PC2: 192.168.10.1
- PC3 and PC4: 192.168.20.1
- Switch 1: 192.168.10.1
- Switch 2: 192.168.20.1

7. Router Configuration

7.1 Router 1 Configuration

The following commands configure Router 1.

```
enable
configure terminal

interface FastEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface Serial0/3/0
ip address 10.0.0.1 255.255.255.252
no shutdown
exit

ip route 192.168.2.0 255.255.255.0 10.0.0.2

end
write memory
```

7.2 Router 2 Configuration

```
enable
configure terminal

interface Serial0/3/0
ip address 10.0.0.2 255.255.255.252
no shutdown
exit

interface FastEthernet 0/0
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

ip route 192.168.1.0 255.255.255.0 10.0.0.1

end
write memory
```

9. Testing and Verification

After configuring the network devices, I used the ping command to test connectivity between the computers.

The ping command uses ICMP Echo Request and Echo Reply messages to check whether another device is reachable.

Testing Commands

On PC1:

```
ping 192.168.20.2
```

On PC4:

```
ping 192.168.10.2
```

Router Verification Commands

```
show ip interface brief
show ip route
show running-config
```

Switch Verification Commands

```
show ip interface brief
show running-config
```

Figure 2: PC1 Ping Test Screenshot

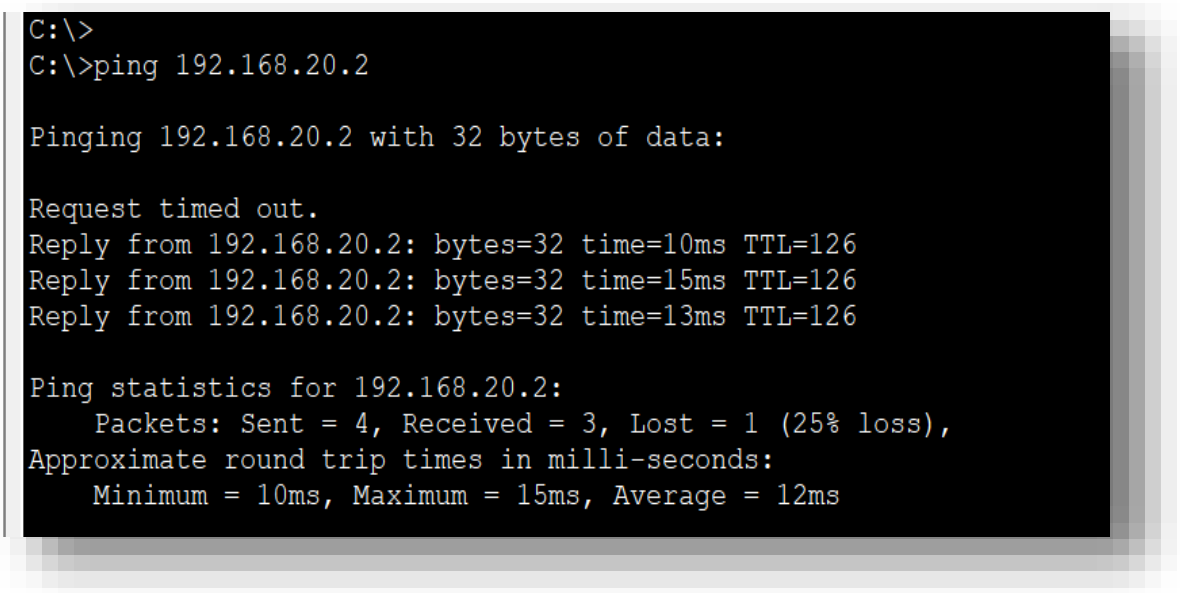


Figure 3: PC4 Ping Test Screenshot

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=126
Reply from 192.168.10.2: bytes=32 time=1ms TTL=126
Reply from 192.168.10.2: bytes=32 time=11ms TTL=126
Reply from 192.168.10.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>
```

Figure 4: Router 1 Verification Output

Show IP Interface Brief

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.10.1    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.1        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Show IP Route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, FastEthernet0/0
L       192.168.10.1/32 is directly connected, FastEthernet0/0
S    192.168.20.0/24 [1/0] via 10.0.0.2
```

Figure 5: Router 2 Verification Output

Show IP Interface Brief

```
R2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        192.168.20.1    YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.2        YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

Show IP Route

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

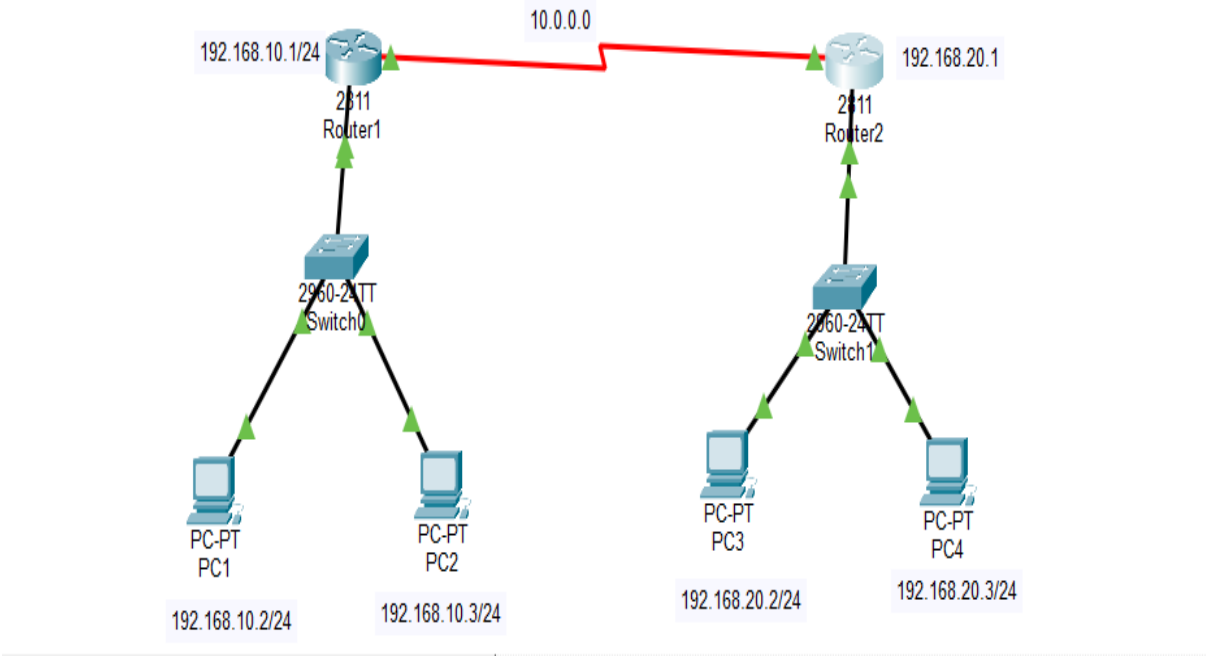
     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
S    192.168.10.0/24 [1/0] via 10.0.0.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, FastEthernet0/0
L       192.168.20.1/32 is directly connected, FastEthernet0/0
```

10. Packet Journey: PC1 to PC4

When PC1 sends a ping request to PC4, the packet follows these steps:

1. PC1 checks whether the destination IP address belongs to its local network.
2. PC4 belongs to a different network, so PC1 sends the packet to its default gateway, Router 1.
3. Switch 1 forwards the Ethernet frame towards Router 1.
4. Router 1 checks its routing table and forwards the packet to Router 2.
5. Router 2 receives the packet and identifies that the destination network is directly connected.
6. Router 2 forwards the packet through Switch 2.
7. Switch 2 delivers the frame to PC4.
8. PC4 receives the ICMP Echo Request and sends an Echo Reply back to PC1 using the reverse path.

Packet Journey Diagram



11. What I Learned

During this practical task, I learned:

- How to install and use Cisco Packet Tracer.
- How to create a network topology using routers, switches, and PCs.
- How to assign IP addresses and subnet masks.
- How to configure router interfaces.
- How to configure static routes between two networks.
- How switches forward frames using MAC addresses.
- How routers forward packets using IP addresses.
- How to use ping to test network connectivity.
- How to use Cisco IOS commands to verify network configurations.

This practical improved my understanding of basic networking concepts and gave me hands-on experience with network configuration.

12. Conclusion

In conclusion, I successfully designed and configured a network topology using Cisco Packet Tracer with two routers, two switches, and four PCs.

I learned how different networking devices work together and how IP addressing and routing allow computers on separate networks to communicate.

The task also helped me understand the journey of an ICMP packet from its source to its destination.

This practical provides a strong foundation for learning advanced CCNA networking concepts.

SUBMITTED FOR:
CCNA Networking Internship

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